June 3, 1963

SUPERSEDING
Fed. Spec. WW-H-171b
February 9, 1959

FEDERAL SPECIFICATION

HANGERS AND SUPPORTS, PIPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pipe hangers, brackets, saddles, clamps, and pipe rolls for all types of piping systems up to 750° F.

1.2 Classification.

1.2.1 *Types.* The hangers and supports shall be types 1 through 52, as indicated herein. Manufacturer's standard products are acceptable if the dimensions of units are at least as large as those shown in table III (see 6.2).

2. APPLICABLE SPECIFICATION, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-P-416—Plating, Cadmium (Electrodeposited).

QQ-Z-325—Zinc-Coating, Electrodeposited, Requirements For.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-12808 Plumbing, Fixtures and Accessories, Packaging of.

MIL-C-14550 Copper Plating, Electrodeposited.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 5340

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standard for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

American Trucking Associations, Inc:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington 6, D.C.)

Official Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York).

American Society For Testing And Materials Specifications:

A 7-61T—Steel for Bridges and Buildings.

A 47-61—Malleable Iron Castings.

A 48-60T—Gray Iron Castings.

A 107-61T—Hot-Rolled Carbon-Steel Bars.

A 120-61T—Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.

A 141-58—Structural Rivet Steel.

A 193-61T—Alloy-Steel Bolting Materials for High-Temperature Service.

A 194-59—Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High Temperature Service.

A 197-47—Cupola Malleable Iron.

A 235-55—Carbon Steel Forgings for General Industrial Use.

A 283-58—Structural Quality, Low and Intermediate Tensile Strength Carbon Steel Plates (Plates 2 inches and Under in Thickness).

A 307-61T—Low-Carbon Steel Externally and Internally Threaded Standard Fasteners.

A 322-57T—Hot-Rolled Alloy Steel Bars.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein and in table I. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product he sells on the commercial market.

Table I. Materials

	Steel	Formed steel	Malleable iron[1]	Cast iron
Hanger Rods--	X	X		
Turnbuckles--	X		X	
Bolts--------	X	X		
Inserts------		X	X	
Beam clamps--	X	X	X	
Pipe clamps--	X	X	X	
Bases--------	X			X
Rollers------	X			X
Brackets-----	X	X	X	
Pipe saddles-	X			X
Protection saddles----	X	X		

[1] See 6.4

3.1.1 *Steel.* Steel and steel products shall conform to one of the following ASTM Specifications: A 7, A 107, A 120, A 141, A 193, A 194, A 235, A 283, A 307, A 322, grades 4130 and 4140.

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3.1.2 *Malleable iron.* Malleable iron shall conform to the requirements of ASTM Specification A 47 or A 197.

3.1.3. *Cast iron.* Cast iron shall conform to the requirements of ASTM Specification A 48.

3.2 **Standard product.** Except where modified herein, it is intended that the hangers and supports and their component parts shall be regular commercial products of the manufacturer or his suppliers. The parts shall be new and unused.

3.3 **Identical items.** All units of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components and accessories. No deviation will be acceptable without prior written approval.

3.4 **Design and fabrication.** Configuration of the pipe hangers and supports shall be similar to the illustrations of types 1 through 52. The design of members not dimensionally detailed herein shall be based on the temperature and weight of the medium transported (testing medium if heavier) and the weight of the piping system including insulation. The allowable stresses for carbon steels used in hangers and supports for pipe systems having temperatures below 650° F. shall not exceed 12,650 p.s.i; for systems having temperatures in the range of 650° to 750° F. shall not exceed 11,150 p.s.i. Carbon steel shall not be used in any member of the hanger or support which contacts the piping system where temperatures exceed 750° F., but may be used in the connecting parts. Adjustable hangers shall be fabricated to permit adjustment after erection while still carrying the load. Standard pipe covering protection saddles, types 40A and 40B, shall be manufactured for thickness of pipe insulation for nominal 1-, 1-1/2-, 2-, 2-1/2-, 3-, 4-, and 5-inch covering. Hangers for N.P.S. or O.D. pipe and copper tubing shall be made to fit the outside diameter of the pipe or tubing and shall meet the test load requirements as listed in table II.

Table II. Hanger Load Requirements

N.P.S.	Nominal Tubing O.D.	Minimum Failure Load Lbs.
3/4 thru 2	1/2 thru 2	750
2 1/2	2 1/2	850
3	3	1050
3 1/2	3 1/2 and 4	1250
4		1500
5		2000
6		2650
8		3800

3.5 **Dimensions for hanger steel.** As indicated on the drawings for types 1 through 52, pipe hangers shall have minimum steel dimensions as listed in table III, or the equivalent cross-sectional area, provided edge distances at boltholes have sufficient load-carrying area. As an exception to this, steel straps adequately plated as outlined in paragraph 3.5.2 and complying with the dimensional requirements of table III after plating will be acceptable. Hanger rod diameters for corresponding pipe sizes are listed in table III column A.

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Table III. Pipe, Hanger and Support Parts Sizes.[1]

Pipe Sizes[2]	A Rod (3)	B	C	D	E (Clevis Top)	F (Clevis Bottom)	G (Clevis Bolt)
1/2	3/8	1/8 x 1	1/4	16g x 7/8			
3/4	3/8	1/8 x 1	1/4	16g x 7/8	1/8 x 1	1/8 x 1	1/4
1	3/8	1/8 x 1	1/4	16g x 7/8	1/8 x 1	1/8 x 1	1/4
1 1/4	3/8	1/8 x 1	3/8	16g x 7/8	1/8 x 1	1/8 x 1	1/4
1 1/2	3/8	1/8 x 1	3/8	16g x 7/8	1/8 x 1 1/8	1/8 x 1	1/4
2	3/8	3/16 x 1 1/4	3/8	14g x 1	1/8 x 1 1/8	1/8 x 1	1/4
2 1/2	1/2	3/16 x 1 1/4	1/2	14g x 1	3/16 x 1 1/4	3/16 x 1 1/4	5/16
3	1/2	3/16 x 1 1/4	1/2	13g x 1	3/16 x 1 1/4	3/16 x 1 1/4	5/16
3 1/2	1/2	3/16 x 1 1/4	1/2	13g x 1	3/16 x 1 1/4	3/16 x 1 1/4	3/8
4	5/8	1/4 x 1 1/4	1/2	12g x 1 1/2	1/4 x 1 1/4	3/16 x 1 1/4	3/8
5	5/8	1/4 x 1 1/4	1/2	12g x 1 1/2	1/4 x 1 1/4	3/16 x 1 1/4	1/2
6	3/4	3/8 x 1 1/2	5/8	10g x 1 1/2	1/4 x 1 1/2	3/16 x 1 1/2	1/2
8	7/8	3/8 x 1 1/2	5/8	10g x 1 1/2	1/4 x 1 3/4	3/16 x 1 3/4	5/8
10	7/8	1/2 x 2	3/4		3/8 x 1 3/4	1/4 x 1 3/4	3/4
12	7/8	1/2 x 2	7/8		3/8 x 2	1/4 x 2	3/4
14	1	1/2 x 2 1/2	7/8		1/2 x 2	1/4 x 2	3/4
16	1	1/2 x 2 1/2	7/8		1/2 x 2 1/2	1/4 x 2 1/2	7/8
18	1 1/8	5/8 x 2 1/2	1		1/2 x 2 1/2	1/4 x 2 1/2	7/8
20	1 1/4	5/8 x 2 1/2	1		5/8 x 3	3/8 x 3	1
24	1 1/2	5/8 x 3	1				

[1]Table III applies only to IPS hangers. For copper tubing hangers, see paragraph 3.5.2.2

[2]All above dimensions are in inches, nominal IPS.

[3]Single rod hangers only.

Stock dimensions of equal cross sectional area are acceptable, provided edges distances at bolt holes have sufficient load carrying area.

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Table III. Pipe, Hanger and Support Parts Sizes. (Cont'd)

Pipe Size[2]	H[4]	J[5]	K	L	M[6]	N	P[7]
1/2			3/16 x 1 1/4			1/8 x 1 1/8	15g x 3/4
3/4			3/16 x 1 1/4			1/8 x 1 1/8	15g x 3/4
1			3/16 x 1 1/4		3/8	1/8 x 1 1/8	16g x 7/8
1 1/4			3/16 x 1 1/4		3/8	1/8 x 1 1/8	16g x 7/8
1 1/2			3/16 x 1 1/4	1/4 x 1 1/4	3/8	1/8 x 1 1/8	13g x 3/4
2			3/16 x 1 1/4	1/4 x 1 1/4	3/8	1/8 x 1 1/8	13g x 3/4
2 1/2		1 1/2	3/16 x 1 1/4	1/4 x 1 1/4	1/2	1/4 x 1 1/4	1/8 x 1 1/8
3		1 1/2	3/16 x 1 1/4	1/4 x 1 1/4	1/2	1/4 x 1 1/4	1/8 x 1 1/8
3 1/2		1 1/2	1/4 x 1 1/4	1/4 x 1 1/4	1/2	1/4 x 1 1/4	1/8 x 1 1/8
4	3	2 1/2	1/4 x 1 1/2	3/8 x 2	5/8	1/4 x 1 1/4	3/16 x 1 1/8
5	3	2 1/2	1/4 x 1 1/2	3/8 x 2	5/8		
6	3	2 1/2	1/4 x 1 1/2	3/8 x 2 1/2	3/4		
8	3	2 1/2	3/8 x 1 1/2	3/8 x 2 1/2	7/8		
10	3	2 1/2	3/8 x 2	1/2 x 2 1/2	7/8		
12	3	2 1/2	1/2 x 2	1/2 x 2 1/2	7/8		
14	4	3	1/2 x 2	5/8 x 3	1		
16	4	3	5/8 x 2 1/2	5/8 x 3	1		
18	4	3 1/2	5/8 x 2 1/2	5/8 x 3	1 1/8		
20	6	3 1/2	5/8 x 2 1/2	3/4 x 3	1 1/4		
24	6	4		3/4 x 3	1 1/2		

[4]Size 26 thru 36H = 6 inches.

[5]Size 30 thru 36J = 4 inches.

[6]Where Type 42 is used on insulated lines, rod sizes may be reduced in accordance with manufacturers catalog recommendations.

[7]Pipe or tubing size.

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3.5.1 *Hanger rod load limitations.* A 25 percent reduction in allowable stresses based on the cross-sectional area at the base of the threads shall be made for threaded members. Hanger rods made from hot-rolled steel rod conforming to ASTM A 107 shall be not less than the sizes listed in table IV for loads and temperatures noted therein.

Table IV. Carrying Capacities of Threaded Hot-rolled Steel Conforming to ASTM A-107.

Nominal rod diameter	Root area of thread	Maximun safe load at rod temperature of 650° F.
Inches	Square Inches	Pounds
3/8	.068	610
1/2	.126	1,130
5/8	.202	1,810
3/4	.302	2,710
7/8	.419	3,770
1	.552	4,960
1 1/8	.693	6,230
1 1/4	.889	8,000
1 3/8	1.053	9,470
1 1/2	1.293	11,630
1 5/8	1.515	13,630
1 3/4	1.744	15,690
1 7/8	2.048	18,430
2	2.292	20,690

3.5.2 *Protective coating.*

3.5.2.1 *Pipe hangers.* When pipe hangers have steel dimensions below those listed in table III, the material shall be electrolytically coated with zinc or cadmium. The coating shall cover the entire surface, and shall be adherent, smooth, continuous, and free from defects such as lumps, blisters, uncoated spots or other defects that may affect the serviceability of the hanger. Cadmium plating shall conform to the requirements of QQ-P-416 (class 3). Zinc plating shall conform to the requirements of QQ-Z-325, (class 3).

3.5.2.2 *Hangers for tubing or brass pipe.* All hangers for copper tubing or brass pipe shall be electrolytically coated steel. The coating shall conform to MIL-C-14550 (class 3). Coated steel hangers for copper tubing need not conform to the dimensional requirements of Table III. Hangers made of solid, compatible, nonferrous metals may be used in place of electrolytically coated steel.

3.6 **Screw threads.** Threads used on pipe hangers and supports shall be unified national-coarse, class 2 female fit, class 1 male fit. The thread for type 39 supports shall be American Standard Pipe Threads. All threads shall conform to National Bureau of Standards Handbook H28.

3.7 **Insulation shields and protection saddles (types 40 and 41).** Type 41 insulation protection shields must be used on insulations having a vapor barrier such as used on refrigeration and air conditioning systems. They must also be used on insulated brass pipe copper tubing lines regardless of the type of insulation. Under no circumstances will the use of type 40A or 40B pipe covering protection saddles be permitted on insulation with a vapor barrier. On non-ferrous insulated lines, type 40A or 40B saddles may be used if suitably modified for both material and method of attachment. Protection shields for low compressive strength refrigeration insulation shall be as follows:

Nominal pipe or tubing size	Length of shield	Gage Number
Inches	*Inches*	
1/2 thru 3-1/2	12	18
4	12	16
5	15	16
6	18	16
8	24	14

NOTE. All shields shall be made for an arc of approximately 180°.

Types 40A and 40B pipe covering protection saddles must be used on insulated steel lines where the use of a roller (types 42 through 48) is warranted.

3.8 **Spring hangers (types 49 thru 52).** At points of support where vertical movement in the pipe line will occur due to expansion or contraction, spring type hangers shall be used. For light loads and where movement is negligible and not critical, types 49 and 50 spring cushions are acceptable. Where movements are in excess of 1/8 inch and do not exceed 3/4 inch, type 51 variable spring hangers are required. On critical systems, any movement in excess of 1/2 inch shall warrant the use of type 52 constant support hangers.

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3.9 Spacing. Recommended hanger spacing shall be as follows:

Size Pipe	Max. Span	Size Pipe	Max. Span	Size Tubing	Max. Span
Inch	*Feet*	*Inch*	*Feet*	*Inch*	*Feet*
1/2 to 1	7	5	16	3/8 through 3/4	5
1-1/2	9	6	17	1	6
2	10	8	19	1-1/4	7
2-1/2	11	10	22	1-1/2 and 2	8
3	12	12	23	2-1/2	9
3-1/2	13	14	25	3	10
4	14	16	27	3-1/2	11
				4	12

The above spacing is based on a combined bending and shear stress of 1500 p.s.i. when the pipe is filled with water and the pitch of the line is such that a sag of 0.1 inch between supports is permissible.

3.10 Finishing. The hangers and supports shall be finished in accordance with the manufacturer's best standard practice.

3.11 Identification marking. The name or initial of the manufacturer, or other mark by which he can readily be identified, shall be legibly cast or die-stamped on a main member of each hanger or support, when practicable.

3.12 Workmanship. The quality of the workmanship shall meet the standards prevalent among manufacturers who normally produce hangers and supports of the type specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification, contract, or order. In the event of conflict, this specification shall govern.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* All hangers or supports of the same type and size offered for delivery at one time shall be considered a lot for the purpose of inspection and test.

4.2.2 *Sampling for examination.* Random samples of hangers or supports shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at Inspection Level II. The AQL shall be 2.5 percent defective.

4.3 Examination. Each of the sample hangers or supports selected in accordance with 4.2.2 shall be visually and dimensionally examined for conformance with the requirements stated in section 3.

4.4 Tests. When specified (see 6.2), samples selected in accordance with 4.2 shall be tested for load carrying capacities as stipulated in table IV, or as more specifically defined in the contract or order. When required, individual component parts shall be

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tested at room temperature on a recognized tensile testing machine. Failure to meet requirements shall be cause for rejection.

4.4.1 When specified in the contract or order the coatings specified in 3.5.2.1 and 3.5.2.2 shall be tested by method 521.1 of Federal Test Method Standard No. 151.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the items shall be subject to inspection to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, Fed. Std. No. 102 should be referred to for the various levels of packaging and packing.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Hangers and supports shall be preserved and packaged in accordance with the applicable level A requirements of MIL-P-12808.

5.1.2 *Level C.* Hangers and supports shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Preserved and packaged hangers and supports shall be packed in accordance with the applicable level A requirements of MIL-P-12808.

5.2.2 *Level B.* Preserved and packaged hangers and supports shall be packed in accordance with the applicable level B requirements of MIL-P-12808.

5.2.3 *Level C.* The hangers and supports shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1. *Civil agencies.* In addition to any special marking required by the contract or order interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 102.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pipe hangers and supports covered by this specification are intended for use with all types of piping systems up to 750° F.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required (1.2.1 and types 1 to 52).

(*c*) Material and diameter of pipe or tubing on which hangers and supports will be used (table III).

(*d*) Name and temperature of medium to be transported when temperature exceeds 650° F.

(*e*) When beam clamps are required give beam size, flange width ("x" dimension), and weight of beam (types 20, 21, and 28 through 31).

(*f*) Thickness of insulation for saddles and, when protection saddles are required, give correct hangers or roller size with which they will be

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used as listed in commercial catalogs (types 40A and 40B).

(*g*) Pipe or tubing size and outside diameter of insulation when type 41 protector is required (3.7).

(*h*) Administrative provisions for inspection (4.1).

(*i*) If load testing in accordance with table IV is required (4.4).

(*j*) Give details of test, other than load testing in accordance with table IV if required (4.4.1).

(*k*) Level of preservation and packaging and level of packing required (5.1 and 5.2).

(*l*) Marking required (5.3).

6.3 This specification conforms in general to the requirements of the current publications of the American Standard ASA B31.1 Codes for Pressure Piping, and to the Manufacturers Standardization Society of the Valve and Fitting Industry, SP-58.

6.4 Military Specification MIL-H-15877, titled Hangers, Pipe (Spring Hangers and Swaybraces, Shipboard Use) covers spring hangers required on shipboard piping systems only. Spring hangers, types 51 and 52 are acceptable for piping systems other than on shipboard where pipeline elevations change due to expansion or contraction of systems (see 3.8).

6.5 Additional types, sizes, etc., of the commodities described herein are commercially available, but those illustrated are considered sufficient for most Federal Government requirements. Malleable iron hangers and supports shall be types or components suitable for the material used and shall be limited to 450° F. Design temperatures for parts of hangers and supports coming in direct contact with pipe and exterior to any insulation may be designed for one-third of the fluid temperature or the ambient temperature, whichever is greater.

6.6 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Pipe hangers, rolls and chairs, not otherwise indicated by name, iron or steel.

Carload minimum weight pounds	40,000
Motor maximum weight pounds	40,000
Motor minimum weight pounds	24,000

Supports—

Pipe supports, iron or steel, consisting of anchors, brackets, rests, or sleeves

Carload minimum weight pounds	36,000
Motor minimum weight pounds	36,000

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

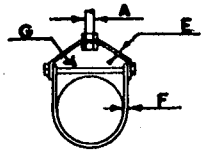
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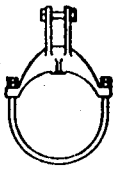
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Air Force MOA

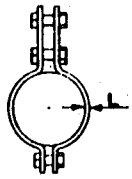
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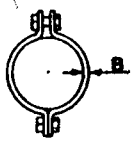
ADJ. STEEL CLEVIS
SIZES 3/4" THRU 20"IPS
TYPE 1



ALLOY STEEL PIPE CLAMP
SIZES 4" THRU 24"IPS
TYPE 2



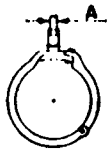
STEEL DOUBLE BOLT
PIPE CLAMP
SIZES 1 1/2" THRU 24"IPS
TYPE 3



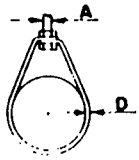
STEEL PIPE CLAMP
SIZES 1 1/2" THRU 24"IPS
TYPE 4



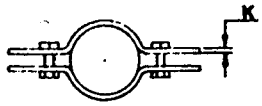
ADJ. MI SWIVEL PIPE RING
SOLID RING TYPE
SIZES 3/4" THRU 8"IPS
TYPE 5



ADJ. MI SWIVEL PIPE RING
SPLIT RING TYPE
SIZES 3/4" THRU 8"IPS
TYPE 6



ADJ STEEL BAND HGR
SIZES 1/2 THRU 8" I.P.S.
3/4 THRU 4" TUB
TYPE 7



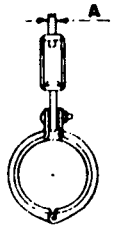
EXT. PIPE OR RISER CLAMP
SIZES 3/4 THRU 20"IPS
1/2 THRU 6" TUB.
TYPE 8



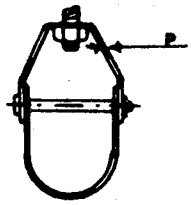
ADJ. BAND HANGER
SIZES 1/2" THRU 8"IPS
1/2" THRU 4"TUB.
TYPE 9



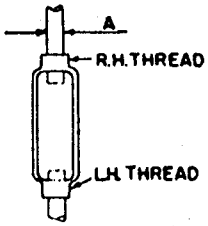
ADJ. SWIVEL RING HGR.
SIZES 1/2" THRU 8"IPS
1/2" THRU 4" TUB.
TYPE 10



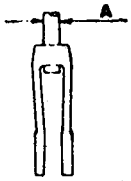
MI SPLIT PIPE RING
(W/WO) TRN BKL ADJ.
SIZES 1/2" THRU 8"IPS
1/2" THRU 4" TUB.
TYPE 11



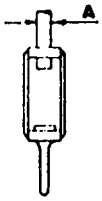
LIGHT DUTY CLEVIS HGR.
SIZE 1/2" THRU 4"
TYPE 12



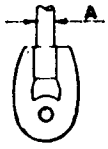
FORGED STEEL
TURNBUCKLE
SIZES 3/8" THRU
1 1/2" ROD
TYPE 13



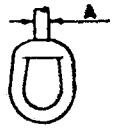
FORGED STEEL
CLEVIS
SIZES 3/8" THRU
1 1/2" ROD
TYPE 14



ADJ. MI TRN BKL
SIZES 3/8" THRU
3/4" ROD
TYPE 15

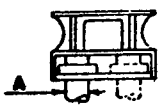


MI SOC ONLY FOR
SPLIT RING HGR.
SIZES 3/8" THRU
7/8" ROD
TYPE 16

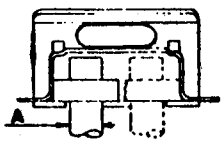


FORGED STEEL WELDLESS
EYE NUT
SIZES 3/8" THRU 1 1/2" ROD
TYPE 17

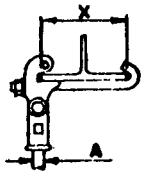
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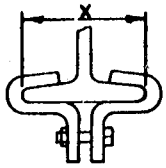
MALLEABLE CONCRETE INSERT
TYPE 18



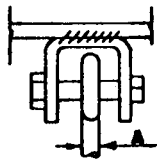
STEEL CONCRETE INSERT
TYPE 19



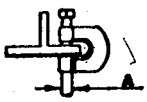
SIDE BEAM CLAMP FOR CHANNEL OR I BEAM
TYPE 20



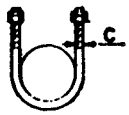
CENTER I BEAM CLAMP
TYPE 21



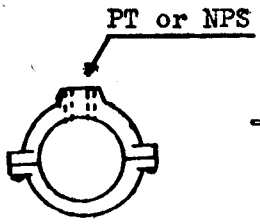
WELDED ATTACHMENT AS SHOWN ON INVERTED LESS BOLT
TYPE 22



MI OR STEEL C-CLAMP
SIZES 3/8" THRU 7/8"
TYPE 23



U-BOLT
1/2" THRU 24"IPS
TYPE 24



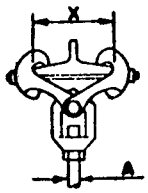
MI EXT. SPLIT RING HGR. 3/8" THRU 4"IPS
1/2" THRU 3" TUBING
TYPE 25



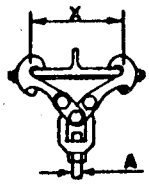
LONG CLIP
3/4" THRU 4"IPS
TYPE 26



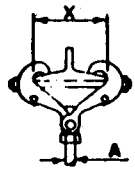
OFFSET J HOOK
1/2" THRU 3"IPS
TYPE 27



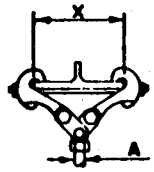
I BEAM CLAMP W/EYE NUT
TYPE 28



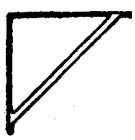
STEEL W. F.
I BEAM W/EYE NUT
TYPE 29



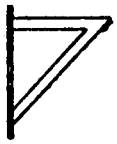
I BEAM CLAMP
W/SWING NUT
TYPE 30



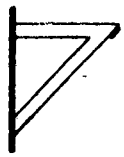
STEEL W.F. I BEAM
W/LINKS & SWING NUT
TYPE 31



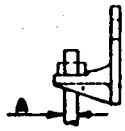
LIGHT WELDED STEEL BRACKET
MAX. LOAD 750 LBS.
TYPE 32



MEDIMUM WELDED STEEL BRACKET
MAX. LOAD 1500 LBS
TYPE 33

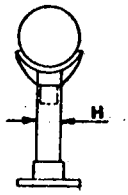


HEAVY WELDED STEEL BRACKET
MAX. LOAD 3000 LBS.
TYPE 34

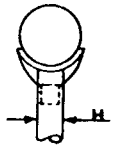


SIDE BEAM BRACKET
TYPE 35

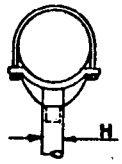
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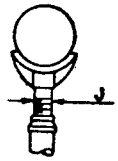
SADDLE SUPPORT WITH BASE
SIZES 4 THRU 36"IPS
TYPE 36



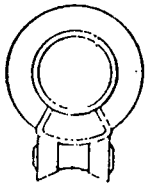
PIPE SADDLE SUPPORT
SIZES 4" THRU 36"IPS
TYPE 37



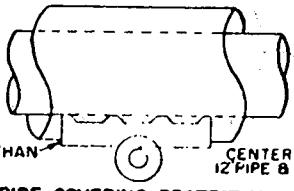
PIPE STANCHION SADDLE
SIZES 4" THRU 36"IPS
TYPE 38



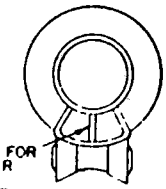
ADJ. PIPE SADDLE SUPPORT
SIZES 2 1/2" THRU 36"
TYPE 39



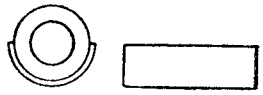
SIZES UP TO AND
INCLUDING 10" IPS
TYPE 40A
(SEE PAR. 3.7)



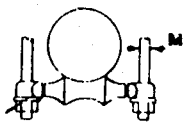
STEEL PIPE COVERING PROTECTION SADDLE
THICKNESS OF METAL
5" I.P.S. DIA. & SMALLER 1/8 MIN.
6" I.P.S. DIA. & LARGER 3/16 MIN.



TYPE 40B
(SEE PAR. 3.7)



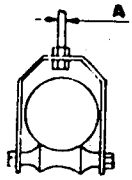
INSULATION PROTECTION
SHIELD
SIZES 3/4" THRU 12"IPS
TYPE 41
(SEE PAR. 3.7)



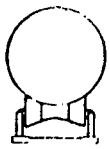
SINGLE PIPE ROLL
SIZES 1" THRU 24"IPS
TYPE 42



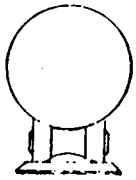
STEEL
ADJ SWIVEL PIPE ROLL
SIZES 2" THRU 12"IPS
TYPE 43



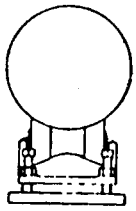
ADJ. ROLLER HANGER
SIZES 1" THRU 20"IPS
TYPE 44



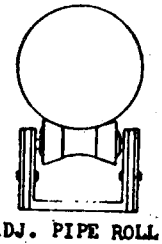
PIPE ROLL COMPLETE
TYPE 45



PIPE ROLL & PLATE
TYPE 46

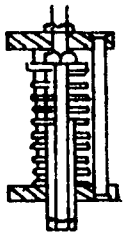


ADJ. PIPE ROLL & BASE
TYPE 47

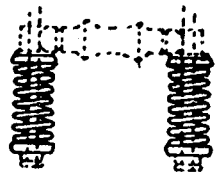


ADJ. PIPE ROLL
TYPE 48

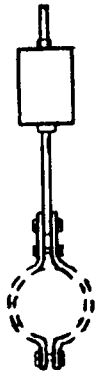
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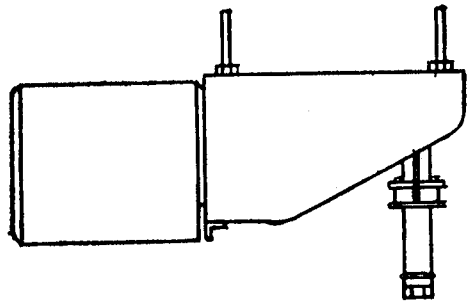
SPRING CUSHION
HANGER
TYPE 49



SPRING CUSHION HANGER
(USE WITH TYPE 42 ROLL)
TYPE 50



VARIABLE SPRING
HANGER
TYPE 51



CONSTANT SUPPORT SPRING
HANGER HORIZONTAL
OR VERTICAL TYPE
TYPE 52

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